

A Traffic Flow Forecasting Model Based on BP Neural Network

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Abstract—Estimation of traffic flow with reasonable accuracy is essential for successful implementation of an Intelligent Transportation System (ITS). Crossroads are important part of urban traffic system, whose flow prediction on each direction is one of the most extraordinary key functions in the urban ITS. Some forecasting models have been developed, but these methods' precision usually can't meet with practical requirement. In this article, a neural network model is presented for forecasting crossroads traffic flow using Backpropagation (BP) Neural Network. Through forecasting traffic flow at Hongqi crossroad in Ganzhou City, the result shows that this model has a considerable accuracy, which provides a new reliable and effective way of forecasting short term traffic flow of crossroads in urban road network.

Index Terms—backpropagation, neural network, traffic flow forecasting, intelligent transportation system, crossroad

I. INTRODUCTION

The rapid growth of the economy has brought great benefits to cities development around the world during the past several decades. However, it also increases the risk of cities environmental problems, such as traffic problem. How to tackle the exiting traffic problem efficiently is one of the most crucial posers in the world. No government can claim that the traffic problem has been settled completely. In the latest of 2000's, experts and scholars in developed countries proposed a concept of Intelligent Transportation System (ITS), which may be the only best way to solve thoroughly the complicated matters. ITS is an entire ground transportation management system which was composed of advanced information technology, data communications transmission technology, electronic sensor technology, electronic control technology and computer processing technology. ITS is a real-time, accurate and efficient integrated transport and management system[1].

After about twenty years research, the development both in theory and practice of ITS is tremendous. In the field of ITS, advanced traffic information system (ATIS) and advanced traffic management system (ATMS) are commonly considered to be the most important and indispensable components of ITS. However, the key of these systems can work efficiently is to achieve real-time and accurate traffic flow forecasts. Traffic flow forecast accuracy will have a direct bearing on traffic

control and vehicle-induced effects. So, the formulation of dynamic traffic flow forecast models is believed to be the key theoretical and technical foundation in establishing real ITS in large-scale road networks.

According to the length of projections time, traffic flow forecasts are divided into long-term traffic flow forecasting and short-term traffic flow forecasting. Short-term traffic flow is less than 15min of traffic flow. With the predicted time shortened, traffic flow changes complexity, randomness and uncertainty. This is the reason that the short-term forecasting of traffic flow is more difficult than long-term forecasting.

Many scholars have been studying on the topic for many years and many forecasting models have been developed. Commonly used methods include average method, ARMA, linear regression, nonparametric regression and so on [2-6]. These methods; forecasting precision usually can't meet with practical requirement. And some complicated methods can't be modeled accurately because traffic flow models of different time section in a day have great diversities. So the choice of suitable traffic flow forecasting method and proper time partition method are crucial to improve traffic flow forecasting precision.

Neural network computing appears to be a promising approach to overcome this shortcoming. A neural network provides a mapping between a set of inputs and corresponding outputs. The network is trained to learn this mapping using a number of training examples. The training is achieved by determining the network's weights. Backpropagation (BP) Neural Network is the most widely used neural network model in civil engineering applications, primarily due to its simplicity.

Crossroads are important part of urban traffic system, whose flow prediction on each direction is one of the most extraordinary key functions in the urban ITS. In this article we present a BP Neural Network model for forecasting the short-term traffic flow on crossroads, which take the influence from the other non-predicted directions and signal timing scheme into account.

II. BP NEURAL NETWORK

The field of neural networks can be thought of as being related to artificial intelligence, machine learning, parallel processing, statistics, and other fields. The attraction of neural networks is that they are best suited to solving the problems

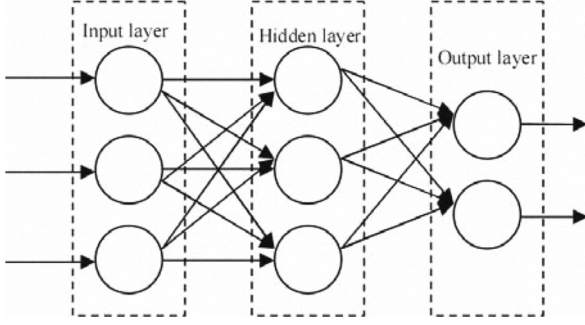


Figure 1. BP Neural Network Topology .

that are the most difficult to solve by traditional computational methods.

The BP Neural Network is a multi-layer network which uses non-linear differentiable functions to do the weight training. It includes the pith of the whole neural network theory. Because of its simple structure and impressive constructiveness, it is widely used in mode identification, function approach, information classification, data condensation, and etc. Its definite math meaning and clear learning algorithm expand its application fields widely.[7]

The simple BP Neural Network architecture is show in Fig.1. Common BP Neural Network has one input layer, some hidden layers and one output layer. Common BP Neural Network has four steps of execution:

1) *Initialization* . We need initialize these three parameters:

$w_{kj}^{(s)}(n)$ is defined as the synaptic weight that corresponds to the connection from neuron unit j in the $(s-1)^{th}$ layer, to k in the s^{th} layer; η is defined as the learning rate and is a constant scaling factor used to control the step size in error correction during each iteration of the backpropagation algorithm; $\theta_k^{(s)}$ is defined as the bias of a neuron. By the way, it can be thought as noise.

2) *Forward Computation*. We can use the following expression to describe it:

$$H_k^{(s)} = \sum_{j=1}^{N_{s-1}} w_{kj}^{(s)} o_j^{(s-1)} + \theta_k^{(s)} \quad (1)$$

$H_k^{(s)}$ is a weighted sum of the k^{th} neuron in the s^{th} layer; $o_j^{(s-1)}$ is neuron output of the j^{th} neuron in the $(s-1)^{th}$ layer.

The computation at the output layer neurons is as follows:

$$o_k^{(s)} = f(H_k^{(s)}) \quad (2)$$

$f(H_k^{(s)})$ is an activation function computed on the weighted sum $H_k^{(s)}$. $f(x)$ is the activation function defined by the following sigmoid function:

$$f(x) = 1 / (1 + e^{-x}) \quad (3)$$

3) *Backward Computation*. In the backward computation step, the weights and biased are updated. Using this step we can reduce the error between the expected value and the output

value in the step 2. It will start from the output layer, and moving back layer by layer, towards the input layer. The local gradients are as follows:

$$\epsilon_k^{(s)} = \begin{cases} t_k - o_k^{(s)} & s = M \\ \sum_{j=1}^{N_{s+1}} w_{kj}^{(s+1)} \delta_j^{(s+1)} & s = 1, \dots, M-1 \end{cases} \quad (4)$$

$\epsilon_k^{(s)}$ is error term for the k^{th} neuron in the s^{th} layer; $\delta_j^{(s+1)}$ is a local gradient for the j^{th} neuron in the $(s+1)^{th}$ layer, t_k is a teaching single. Get a local gradient from the equation:

$$\delta_k^{(s)} = \epsilon_k^{(s)} f'(H_k^{(s)}) \quad s = 1, \dots, M \quad (5)$$

$f'(H_k^{(s)})$ is the derivative of the activation function.

Then the weight (and bias) change is calculated for all the weights as follows:

$$\Delta w_{kj}^{(s)} = \eta \delta_k^{(s)} o_j^{(s-1)} \quad k = 1, \dots, N_s \quad j = 1, \dots, N_{s-1} \quad (6)$$

$\Delta w_{kj}^{(s)}$ is the change in synaptic weight (or bias)

corresponding to the gradient of error for connection from the j^{th} neuron in the $(s-1)^{th}$ layer, to the neuron k in the s^{th} layer.

From equations (3)(4)(5), we can rewrite this equations:

$$\Delta w_{kj}^{(s)} = \begin{cases} \eta (t_k - o_k^{(s)}) o_j^{(s-1)} (1 - o_k^{(s)}) & s = M \\ \left(\sum_{j=1}^{N_{s+1}} w_{kj}^{(s+1)} \delta_j^{(s+1)} \right) o_k^{(s)} (1 - o_k^{(s)}) & s = 1, \dots, M-1 \end{cases} \quad (7)$$

$$k = 1, \dots, N_s \quad j = 1, \dots, N_{s-1}$$

Finally, all the weights (and biases) are updated.

4) *Iteration*. Repeat the Forward and Backward Computations for each training example. When training is completed, carry out the Forward Computation.

III. MODEL CONSTRUCTION

A. A Brief Introduction to Crossroad Traffic Problem

Crossroad is an important part of urban transportation. As the simplified model shown in Figure 2, how to maximize the crossroad traffic flow and minimize the waiting time of vehicles in each direction is crucial to improving the traffic volume and efficiency of the whole transport network. Optimizing the time-sharing scheme of traffic light is the current uppermost way to improve crossroad traffic flow and reduce the average latency.

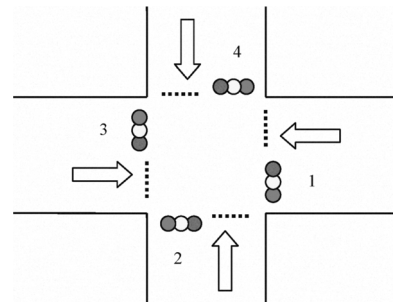


Figure 2. Crossroad Model.

There are many ways of optimization, among which Automatic Induction Method is applied in ITS. At a fixed or non-fixed cycle, intersection traffic light turns bright green light to release waiting vehicles in all directions according to a certain order, which is repeated again and again, so that vehicles can pass the intersection in an orderly manner. As shown in Figure 2, vehicles in 1 and 3 directions and that in 2 and 4 directions are released by turns. Time cycles of releasing vehicles in each direction are fixed in a fixed time-sharing scheme of traffic light. If the green light time in 1 and 3 directions is one minute, the same as in 2 and 4 directions, a cycle is 2 minutes. In Automatic Induction Method, there are maximum inductive green (the longest time of green light in inductive mode), minimum inductive green (the shortest time of green light in inductive mode), and additional green (automatically added time for green light) in each direction. If detector finds no vehicle in a certain time interval, the traffic light in 1 and 3 directions turn from green to red when green light time here exceeds minimum inductive green. However, if detector finds vehicles in a certain time, there will be an additional green until maximum green ends, and then vehicle in 2 and 4 directions are released. Traffic light in ITS works according to fixed time-sharing scheme and auto-inductive time-sharing scheme, which is preset by time-sharing scheme in database. Such as peak hours in one day are 08:00-10:00, 13:00-15:00 in a fixed time-sharing scheme, other working time in automatic inductive time-sharing scheme should be as follows. Fixed time-sharing scheme varies according to different time distributed in each direction in a cycle, while automatic inductive time-sharing scheme varies according to various maximum inductive green, minimum inductive green and additional green in each direction.

B. BP Neural Network Construction

Since detectors in the ITS send the traffic flow information to the central database every 5 minutes, we should design a BP Neural Network to forecast the traffic flow in each direction in the following 5 minutes. Considering one day as a period, and according to different timing plans, we set up models by BP Neural Networks to forecast the traffic flow in each direction for different timing plans. Clearly the traffic flow in one direction is influenced by that of the other three, and traffic flow in two crossing directions will affect the forecast even more, especially under the induction mode.

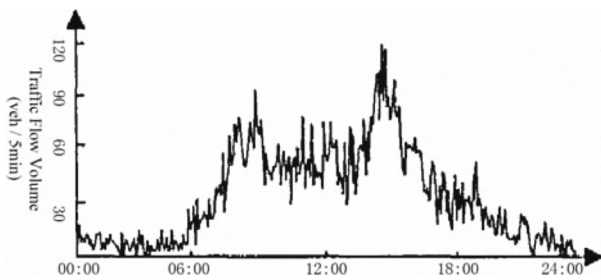


Figure 3. Traffic Flow Volume.

Here $g_1(t+1)$ is used to indicate the traffic flow in direction 1 in the following 5 minutes.

$$g_1(t+1) = Y_1(g_1(t), g_1(t-1), g_1(t-2), g_2(t), g_3(t), g_4(t)) \quad (8)$$

$g_1(t)$ 、 $g_1(t-1)$ 、 $g_1(t-2)$ mean the traffic flow in forecasting direction in the previous three times respectively; $g_2(t)$ 、 $g_3(t)$ 、 $g_4(t)$ mean the traffic flow in the other three directions in the previous time respectively. So Y_2 、 Y_3 、 Y_4 , functions of the forecasting traffic flow in the other directions can be achieved for the same reason. For each Y function a BP Neural Network Approximation is conducted.

The six values, that is, the traffic flow in forecasting direction in the previous three times respectively, and the traffic flow in the other three directions in the previous time respectively, is given as input of the BP Neural Network; and traffic flow in the next time is given as the output.

Since according to Kolmogorov's theorem, any continuous function can be accurately achieved by a three-layer feed forward network, three layers of network are used here, including the input layer, the Single-Sigmoid hidden layer and Linear output layer. The number of neurons on the input layer is calculate $2 \times 6 + 1 = 13$ according to Kolmogorov's theorem, so number of neurons on the hidden layer is chosen as 15 here to increase the accuracy of forecasting and the stability of convergence.

IV. EXAMPLE

Part of the data in this experiment comes from the historical data on Hongqi crossroad in Ganzhou City from June 20th to July 25th in 2008. What is shown in Figure 3 is the traffic flow volume from west to east on Hongqi crossroad on June 20th. This direction is also chosen as the forecasting direction and the other three directions as auxiliary input conditions forecast. The traffic flow data fluctuate greatly due to the Signal cycle and 5-minutes sampling, which is the characteristics of traffic flow on crossroads.

Matlab 7.0 software and BP Neural Network tool box is applied here, with Levenberg-Marquardt as the training method for its applicability in function Approximation.

The timing plans on this crossroad can be divided into five periods, from which three periods are drawn, that is 08:00-10:00, 10:00-14:00 and 14:00-16:00. They are also times that need to be analyzed most. In the experiment the time from June 20th to July 19th is drawn as training sample and from July 20th to 25th as samples for assessment; the cycle index is 200, 300 and 400 respectively, and error limit is 0.001.

The experiment data are shown in Table 1, and the outcome is shown in Table 2.

The outcome of the experiment indicates that BP Neural Network is stable on the whole after 300 cycle training without clear accuracy change. But the time for training is in proportional to the training index; while the accuracy of the forecast can reach 85 percent, which has achieved the effects of assisting traffic and optimizing timing plan.

TABLE I. EXPERIMENTAL DATA

<i>Time</i>	<i>Training Samples</i>	<i>Evaluating Samples</i>
08:00—10:00	573	114
10:00—14:00	4116	693
14:00—16:00	1029	187

TABLE II. EXPERIMENTAL RESULTS

<i>Time</i>	<i>Cycle Training Times</i>	<i>Accuracy</i>	<i>Spending Time of Training</i>	<i>Spending Time of Evaluating</i>
08:00 — 10:00	200	85.13%	6.26	0.0051
	300	86.58%	10.15	0.0074
	400	85.77%	14.31	0.0109
10:00 — 14:00	200	85.72%	43.57	0.0093
	300	87.14%	66.49	0.0148
	400	86.30%	89.17	0.0187
14:00 — 16:00	200	86.96%	13.24	0.0079
	300	88.34%	20.83	0.0126
	400	87.75%	27.40	0.0135

V. CONCLUSIONS

With the foundation of sufficiently analyzing the BP neural network and the characteristic of the prediction of traffic flow,

this article put forward a new means to predict the traffic flow on the base of BP Neural Network. This experiment has taken the influence from other direction ,which are not being predicted, and the influence from traffic signal time assignment on the prediction of traffic flow into consideration. The result of the experiment basing on realistic data is satisfactory. From now on, we will make deeper research on the characteristics of crossroad traffic and other analogous neural network and choose better structure and type of network to attain more accurate result of prediction.

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